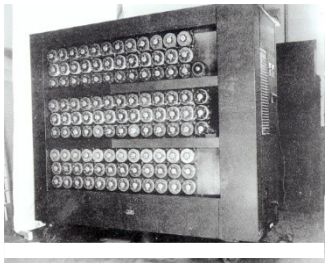
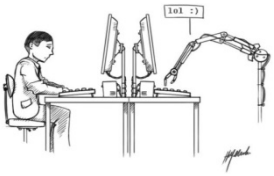
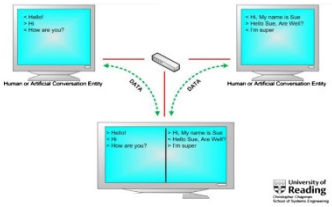
# Individual Task 01: Timeline of Major Milestones in Artificial Intelligence

### 1) Foundations and Early Development (1940–1970)






**1943 – Artificial Neuron Concept**

Warren McCulloch and Walter Pitts proposed the first mathematical model of an artificial neuron. They demonstrated that biological brain functions could be represented using logical and mathematical structures. This work laid the theoretical foundation for neural networks and computational intelligence.

**1950 – Turing Test**

Alan Turing introduced the Turing Test in his paper *"Computing Machinery and Intelligence."* The test evaluates whether a machine can exhibit intelligent behavior indistinguishable from that of a human. It became a foundational philosophical and practical benchmark in AI research.

**1956 – Birth of Artificial Intelligence**

The term "Artificial Intelligence" was officially coined at the Dartmouth Conference, organized by John McCarthy and other researchers. This event marked the formal beginning of AI as a scientific discipline.

**1960s – Early AI Programs**

During the 1960s, AI research focused on symbolic and rule-based systems. Early developments included game-playing programs and theorem-proving systems. These systems relied on logical reasoning and predefined rules to solve structured problems.

**Key Contributions and Limitations of Early AI**

**Contributions:**

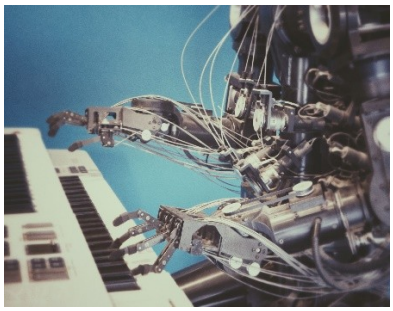
- Introduced the concept of machine intelligence
- Established formal evaluation methods (e.g., Turing Test)
- Developed logical and symbolic reasoning frameworks

**Limitations:**

- Strong dependence on hard-coded rules
- No capability to learn from new data
- Limited computational power

**Key Insight:**
This early phase established the theoretical foundations of AI. However, systems were rigid and suitable only for controlled environments due to limited adaptability and processing capacity.

## 2) AI Winter, Expert Systems, and Machine Learning Shift (1970–2000)



### 1974–1980 – First AI Winter

AI research faced a major decline in funding and interest because of unrealistic expectations and insufficient computing power. This period became known as the First AI Winter.

### 1980s – Expert Systems Era

AI regained popularity through Expert Systems, which simulated human expertise in specific domains using rule-based logic.
**Example:** XCON, used for configuring computer systems in industry.

### Late 1980s – Second AI Winter

Although initially successful, expert systems were expensive to build and maintain. Their inability to scale or adapt led to another decline in funding, known as the Second AI Winter.

### 1997 – Deep Blue vs. Human Intelligence

IBM’s Deep Blue defeated world chess champion Garry Kasparov. This milestone demonstrated that machines could outperform humans in complex strategic and decision-based tasks.

### Late 1990s – Rise of Machine Learning

AI research shifted from manually programmed rules to Machine Learning (ML). In ML, systems learn patterns directly from data using statistical techniques, enabling better adaptability and performance in real-world problems.

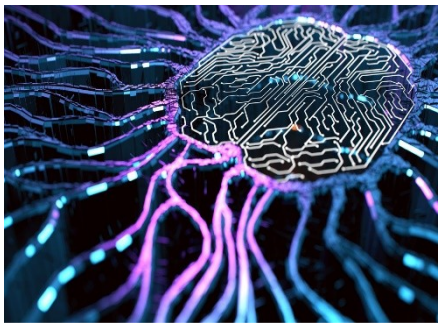
### Why Expert Systems Failed

- Required continuous manual rule updates
- Difficult to scale for complex problems
- Unable to learn from new data
- High development and maintenance costs

### Technical Limitation

Rule-based systems lacked flexibility and efficiency for large-scale, dynamic environments. These limitations led to the adoption of data-driven Machine Learning approaches that could automatically adapt and improve with experience

### 3) Deep Learning and Modern AI Era (2000–Present)



**2000s – Data-Driven AI**

The growth of the internet and big data significantly enhanced AI capabilities. AI systems improved in applications such as search engines, speech recognition, and recommendation systems by leveraging large datasets and computational power.

**2012 – Deep Learning Breakthrough**

Deep neural networks achieved record-breaking performance in image recognition tasks, particularly in the ImageNet competition. This event marked the beginning of the Deep Learning revolution and accelerated research in neural networks.

**2016 – AlphaGo Achievement**

DeepMind’s AlphaGo defeated Go world champion Lee Sedol. This achievement highlighted the power of deep reinforcement learning and demonstrated AI’s capability in strategic reasoning beyond brute-force computation.

**2020s – Generative and Responsible AI**

Modern AI systems can generate text, images, and code, and are widely applied in healthcare, education, finance, and industry. Technologies such as voice assistants and chatbots have become common in daily life.

However, rapid advancements have raised concerns regarding bias, fairness, privacy, transparency, and safety. As a result, responsible and ethical AI development has become a major global focus.

**Impact on Society**

AI has transformed industries by enabling automation, predictive analytics, and intelligent decision-making. It supports medical diagnosis, autonomous systems, financial forecasting, and personalized learning. At the same time, governments and organizations emphasize ethical standards and regulatory frameworks to ensure safe deployment.

**Conclusion**

The history of Artificial Intelligence shows a clear progression from symbolic reasoning and rule-based systems to data-driven Machine Learning and Deep Learning models. Each stage—early theoretical foundations, expert systems, statistical learning, and modern generative AI—contributed essential advancements that shaped today’s intelligent technologies.Currently, AI development focuses not only on performance and scalability but also on ethical responsibility, transparency, and sustainability. As technology continues to evolve, AI is expected to play an increasingly significant role in shaping the future of society.